

30-10-15(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov – Dec 2015

Max. Marks:100

Class: S.E.

Name of the Course: Data Structures

Course Code: UCEC303

Duration: 3.00Hrs.

Semester: III

Branch: Computer

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	(i) The most appropriate matching for the following pairs X: depth first search 1. Hash table Y: breadth-first search 2: queue Z: search 3. Stack (a) X—1 Y—2 Z - 3 (b) X—3 Y—1 Z - 2 (c) X—3 Y—2 Z- 1 (d) X—2 Y—3 Z - 1 (ii) The following numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree? (a) 2 (b) 3 (c) 4 (d) 6 (iii) Postorder traversal of a given binary search tree, T produces the following sequence of keys 10, 9, 23, 22, 27, 25, 15, 50, 95, 60, 40, 29. Which one of the following sequences of keys can be the result of an in-order traversal of the tree T? a) 9, 10, 15, 22, 23, 25, 27, 29, 40, 50, 60, 95 b) 9, 10, 15, 22, 40, 50, 60, 95, 23, 25, 27, 29 c) 29, 15, 9, 10, 25, 22, 23, 27, 40, 60, 50, 95 d) 95, 50, 60, 40, 27, 23, 22, 25, 10, 9, 15, 29 (iv) A single array A[1..MAXSIZE] is used to implement two stacks. The two stacks grow from opposite ends of the array. Variables top1 and top2 (top1 < top2) point to the location of the topmost element in each of the stacks. If the space is to be used efficiently, the condition for "stack full" is a) (top1 = MAXSIZE/2) and (top2 = MAXSIZE/2+1) b) top1 + top2 = MAXSIZE c) (top1 = MAXSIZE/2) or (top2 = MAXSIZE) d) top1 = top2 - 1	2 2 2 2

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	(v) The Breadth First Search algorithm has been implemented using the queue data structure. One possible order of visiting the nodes of the following graph is <pre> graph TD M --- R M --- Q N --- M N --- Q O --- N O --- P Q --- P </pre> (a) MNOPQR (b) NQMPOR (c) QMNPRO (d) QMNPOR	2
Q 1 (b)	What is an AVL tree? Construct AVL tree for the following data. Mention the type of rotation for each case. 50, 25, 10, 5, 7, 3, 30, 20, 8, 15	10
Q2 (a)	Write algorithm for evaluation of postfix Expression using tree OR Write a C program to implement infix to postfix conversion	10 10
Q2 (b)	What is selection sort ? Write a C program to implement it. Show o/p after every pass for the following numbers 10 , 30 ,20 ,40 , 50 OR What is Hashing? What do you mean by collision use linear probing and mod division to store the values in Hash table of SIZE 10. The numbers are 10 , 15 , 13 ,25 , 27 , 37 , 54 , 65 , 63	10 10
Q3 (a)	Write a C program to search a given number in Doubly linked list ? OR With reference to Singly linked list write a C program to sort the linked list in ascending order	10 10
Q3 (b)	Explain multiplication of sparse matrix. OR Suggest data structure to represent polynomial with 3 variables and write a function to add two polynomials	10 10
Q4 (a)	What do you mean by Recursion ? State the general guideline for converting any recursive function to non-recursive function with example	10
Q4 (b)	Explain Queue as an ADT ? What are the different types of QUEUE.	10
Q5 (a)	Write a program in C to check whether the given string is Balanced Parenthesized or not ?	10
Q5 (b)	What is Binary threaded search tree ? Write a C declaration for it. Show insertion of following numbers in Binary threaded search tree. : 10 , 20 , 15 20 , 25 , 35 , 40	10